

N-Grams

Bigrams, trigrams, and how word combinations add context to bag of words.

Opposite Sentences, Almost the Same Vector

DATASET

S1 ~~The~~ room ~~is~~ clean
S2 ~~The~~ room ~~is~~ **not** clean

AFTER STOPWORDS

S1 room clean
S2 room **not** clean
"not" is kept on purpose: it flips the meaning.

	room	not	clean
S1	1	0	1
S2	1	1	1

Only **1 value** differs. Cosine similarity $\approx$ **0.82**, so they look similar.

In reality the two reviews mean the opposite.

DEFINITION

N-gram = n Words Next to Each Other

S2 CLEANED room not clean

Unigram

n = 1

room

not

clean

Bigram

n = 2

room not

not clean

Trigram

n = 3

room not clean

STEP 1

Add Bigrams to the Vocabulary

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S2 ~~The~~ room ~~is~~ **not** clean

AFTER STOPWORDS

S1 room clean

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"not" is kept on purpose: it flips the meaning.

UNIGRAMS

room

not

clean

BIGRAMS

S1

room clean

S2

room not

not clean

3 unigrams + 3 bigrams = **6 features**

Unigrams + Bigrams: Now the Vectors Differ

DATASET

S1 ~~The room is~~ clean
S2 ~~The room is~~ **not** clean

AFTER STOPWORDS

S1 room clean
S2 room **not** clean

"not" is kept on purpose: it flips the meaning.

	room	not	clean	room clean	room not	not clean
S1	1	0	1	1	0	0
S2	1	1	1	0	1	1

UNIGRAMS ONLY
1 difference · similarity 0.82

WITH BIGRAMS
4 differences · similarity 0.52

Before and After Bigrams

DATASET

S1 ~~The~~ room ~~is~~ clean

S2 ~~The~~ room ~~is~~ **not** clean

AFTER STOPWORDS

S1 room clean

S2 room **not** clean

"not" is kept on purpose: it flips the meaning.

UNIGRAMS · [ROOM, NOT, CLEAN]

S1 = [**1**, 0, **1**]

S2 = [**1**, **1**, **1**]

UNIGRAMS + BIGRAMS · 6 FEATURES

S1 = [**1**, 0, **1**, **1**, 0, 0]

S2 = [**1**, **1**, **1**, 0, **1**, **1**]

ngram_range in scikit-learn

```
CountVectorizer(ngram_range=(min_n, max_n))
```

NGRAM_RANGE	USES	FEATURES FOR S2 "ROOM NOT CLEAN"
(1, 1)	Unigrams	room, not, clean
(1, 2)	Unigrams + bigrams	room, not, clean, room not, not clean
(1, 3)	Unigrams + bigrams + trigrams	... + room not clean
(2, 3)	Bigrams + trigrams	room not, not clean, room not clean

What to Remember

- 01 Single words miss context: "room clean" and "room not clean" look similar.
- 02 N-gram = n neighbouring words. Bigram = 2, trigram = 3.
- 03 Adding n-grams as features makes opposite sentences clearly different.
- 04 scikit-learn: `ngram_range=(1, 2)`. Works with BoW and TF-IDF.